1. Consider your lab experiences and any relevant equations to fill in the blank with the appropriate answers.

A. A sample was determined to contain 1.03356 µg analyte/mL with a 95% confidence limit of 0.000821 µg /mL. Report the concentration to the correct number of significant figures based on the confidence limit. _____1.0336 µg/mL ________

B. In a mixture of two solutes where one is polar (A) and the other is non-polar (B), which solute will elute first on a C18 column, A or B? _____ A ____

C. During quantitative chemical analysis experiments, you are usually asked to record replicate measurements. What type of error is minimized by repeat measurement, random or systematic error? _______random________

D. During the analysis of fluoride content in mouthwash using an ion-selective electrode, it was important to adjust the pH of the solution using the TISAB buffer. If the pH of the solution is too high would the determined fluoride content be lower or higher than the true value? __higher (OH^- interference leads to an overestimated F-)__________

E. Which calibration curve is expected to show a non-zero and positive y-intercept, external calibration curve or standard addition calibration curve? _____ (standard addition calibration because the zero standard added solution contains analyte from the unknown) _________

2. A calibrated curve based on a set of known standards was used to measure the μg RNA in an unknown. The responses (absorbances) from the standards and unknown were measured and reported here. The first two lines from the Linest output are also provided here.

	μg RNA	Absorbance (260nm)
Standard1	0	0.003
Standard2	11.02	0.24
Standard3	21.95	0.44
Standard4	34.15	0.71
Unknown – run 1		**0.33**
Unknown – run 2		**0.31**

Linest (first two lines)	
0.0205	0.00454
0.00051	0.0107

The expression for calculating the standard error in the μg RNA in the unknown is given here

$$s_c = \frac{s_r}{|m|}\sqrt{\frac{1}{M} + \frac{1}{N} + \left(\frac{(\bar{y}_u - \bar{y})s_m}{ms_r}\right)^2}$$

What numerical values of $\bar{y}$, $\bar{y}_u$, M and N should be used in this equation?

$\bar{y}$	= (0.003+0.24+0.44+0.71)/4=0.35
$\bar{y}_u$	=(0.33+0.31)/2 = 0.32
N:	= 4 number of standards used to generate the calibration curve)
M:	=2 number of unknown readings

3. Consider the provided data from the calibration of a 10mL volumetric pipet. The volumes of dispensed water were calculated from the mass using the density of water at the lab temperature. The average and standard deviation were calculated using the Average() and stdev() functions in Excel.

	Mass of dispensed water (g)	Actual volume of dispensed water (mL)
Trial 1	9.9748	9.994833
Trial 2	9.9801	10.00014
Trial 3	9.9795	9.999542
	Average	9.99817
	Standard deviation	0.002908

a. Report the 95% confidence interval in the average volume delivered by this pipet. Report the average volume delivered to the correct number of significant figures based on the confidence limit.

Average = 9.99817 mL

$S_{average} = 0.002908 * \sqrt{\frac{1}{3}} = 0.001679..\text{mL}$

$CL_{average} = t * S_{average} = 4.303 \times 0.001679.. = 0.00722\ \text{mL}$

$n = 3$
$df = 2$
$\Rightarrow t_{table} = 4.303$

Confidence interval $= 9.998 \pm 0.007$ mL

b. Is the determined average volume significantly different than the label volume (10mL) at the 95% confident level? Support your answer with the appropriate calculations/equations for credit.

two possible ways to answer:

1) confidence interval (from part a.) includes 10 mL

OR

2) $t_{calc} = \frac{|9.99817 - 10|}{0.001679..} = 1.08 < t_{table} = 4.3$

both lead to the conclusion:
the determined average volume is NOT significantly different than the Label at the 95% Confidence Level

4. *Consider the titration of 25.00 mL of a solution of unknown KHP concentration with a 0.1295 M NaOH secondary standard solution. Phenolphthalein is used to identify the endpoint of the titration. Three titration attempts were performed and the average volume of NaOH needed to reach the endpoint is found to be 21.25 mL*

a. *Calculate the molarity of the KHP solution. Report the answer to 4 significant figures.*

1:1 acid:base stoichiometry $\Rightarrow M_{KHP} \times V_{KHP} = M_{NaOH} \cdot \bar{V}_{NaOH}$

$\Rightarrow M_{KHP} \times 25 = 0.1295 \times 21.25$

$\Rightarrow M_{KHP} = 0.1101$ M

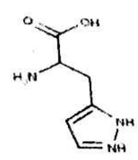
b. *If the buret was rinsed with DI water and was not conditioned with the NaOH solution before filling with NaOH, do you expect the calculated concentration of KHP from this titration to be higher or lower than the expected value? Very briefly explain how you reached your answer for credit.*

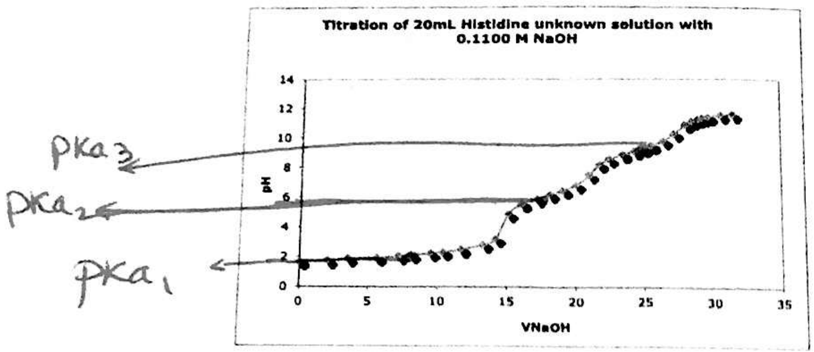
DI left behind in buret $\Rightarrow$ NaOH standard is diluted and its actual concentration is less than the labeled ('true') value.

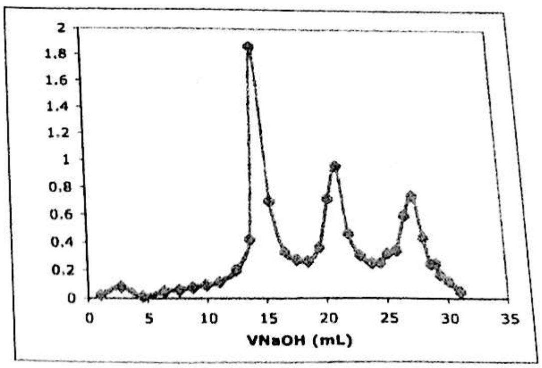
$\Rightarrow$ more NaOH will be needed to reach the end point $\Rightarrow \bar{V}_{OH}$ will be higher and the M_{KHP} will be higher than the expected ('true') value

5. Use the provided titration curve and first derivative plot to answer the following questions about the titrated L-Histidine hydrochloride solution.

(the COOH proton is the most acidic, then the $-NH^{1+}$, then the $-NH^{3+}$ protons)







a. Estimate the pKa_2 of Histidine. Explain/mark on the curve above how you reached your answer.

$pKa_2 \sim 5.9-6$ (halfway between the 1st & 2nd end points)

b. What do the three maxima correspond to in the plot (right) of $\Delta pH/\Delta V$ vs V_{NaOH}?

they correspond to the volumes of NaOH at the 1st, 2nd, and 3rd end points of the titration

6. Cell potential readings in mV were recorded when a F^- ion-selective electrode was immersed in a series of standard solutions to which TISAB buffer was already added. The plot of E (in mV) vs log $[F^-]$ was linear. Regression statistics were determined using LINEST function and excel (given here).

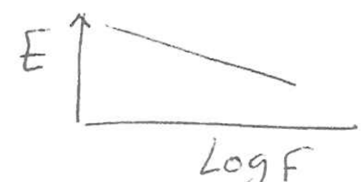
m ←	-54.6192	50.4437	→ b
Sm ←	0.51776	1.37588	→ Sb
	0.99955	1.36997	→ Sy
	11128.03	5	→ df
	20885.50	9.38418	

Know these outputs of the Linest function

→ t = 2.57 from t-table

a. A mouthwash sample with unknown F^- concentration was also analyzed after addition of TISAB buffer and gave a reading of 210.5 mV.

The s_c expression given in the equation sheet was used to determine the standard error in the log$[F^-]$ for the unknown and returned 0.0250. Calculate the 95% confidence interval in the molar concentration $[F^-]$ in the unknown. Report the [F-] in the unknown to the correct number of significant figures based on the confidence limit.



$$E = m \, \text{Log}[F^-] + b$$

⇒ for the unknown:

$$210.5 = -54.6192 \, \text{Log}[F^-] + 50.4437$$

$$\Rightarrow \text{Log}[F^-] = \frac{(210.5 - 50.4437)}{(-54.6192)} = -2.9304\ldots$$

$$[F^-] = 10^{-2.9304\ldots} = 0.001174$$

Calculate $S_{[F^-]}$ then $CL_{[F^-]}$:

$$S_c = S_{\log[F^-]} = 0.0250 \Rightarrow \text{propagation of error: } S_{[F^-]} = 2.303 \times 0.0250 \times 0.001174 = 6.759 \times 10^{-5}$$

see equation sheet!

$$CL_{[F^-]} = t \times S_{[F^-]} = 2.57 \times 6.759 \times 10^{-5}$$

$$\Rightarrow CL_{[F^-]} = 0.000174\ldots$$

⇒ 95% Confidence interval in $[F^-]$: $0.0012 \pm 0.0001_{74}$ M